

48V Electrical Architecture

Document role

- This file owns the final 48V electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in docs/implementation/ .
- Keep unresolved vendor gates, risk state, and follow-up closure items in docs/core/TRACKING.md .
- Keep broad project sequencing and day-to-day execution framing in docs/core/PROJECT.md or the active plan docs.

As-of date: 2026-08-12

Purpose: hold the finalized, concise 48V house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- [Electrical topology diagram](#)
- [Electrical fuse schedule](#)
- [Mechman / WS500 / APM-48 install guide](#)
- [Systems](#)
- [Tracking](#)
- [Estimated BOM](#)

Final direction

- House architecture stays 48V core with 3x 51.2V 100Ah batteries in parallel.
- Alternator charging is the dedicated 48V secondary alternator path; obsolete pre-Mechman charger hardware is removed from active planning.
- Protection baseline is Mechman + WS500 + Balmar APM-48 .
- Manual alternator-charge enable/disable is through Ford Upfitter Switch #3 .
- Upfitter #3 is a low-current control signal only. It does not carry alternator output current.
- WS500 white Feature-In is reserved for future fault-interlock work, not required in Phase 1.

Current commissioning state (2026-08-02)

- 48V bus has been live-tested: owner measured 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II DC/inverter mode has been switched on with inverter light illuminated, slight normal hum, and no reported error lights.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect.
- Cerbo GX access point/remote-console workflow is active; Cerbo power is a small inline fused feed from the 48V system side and MultiPlus communication is via VE.Bus RJ45.
- Shore charging has been tested through the MultiPlus at household-outlet current limits. The first short test proved basic AC-in/charger function; later owner verification redid the settings and confirmed first-battery behavior entered bulk, then quickly transitioned to absorption at/near 100% as planned.
- MultiPlus LiFePO4 charge-profile programming/verification is treated as closed for the current shore-charger setup. Current target is 56.8V absorption/charge, 54.0V float, short absorption dwell, equalization off, conservative LiFePO4 storage behavior, and source-current limit matched to the actual shore circuit. The Dumfume manual's 58.4V +/-0.2V value is documented, but because the same manual also lists 58.4V as charge-limit/over-charge protection voltage, it is not the active routine charger target. DVCC remains disabled unless a documented BMS/GX control path is added.
- The electrical module is hard-mounted through the finished floor to registered truck-bed hardpoints and is now tied into the hard-mounted Bench/Galley extrusion structure; owner reports the integrated assembly is extremely stiff. The remaining road-restraint work is battery/cooler capture plus terminal/cable protection. The driver-rear shore inlet and cable route are physically installed. One accessible, enclosed, conductor/gauge-rated three-wire L/N/PE splice into the AC-input side remains before dead checks and first use.
- All three batteries' 2/0 AWG branch leads are cut, lugged, heat-shrunk, and landed at the battery-side positive/negative busbars. The three batteries are not yet paralleled: Battery 1 completed the corrected isolated charge cycle, Battery 2 remained in normal upper- 54V bulk at the overnight 2026-08-02/03 checkpoint, and Battery 3 remains pending before individual rest/record and <=0.1V matching.
- Owner reports the Orion now charges the 12V buffer battery correctly and the 12V system is operating as intended. Remaining commissioning gates are the shore L/N/PE splice plus polarity/PE/neutral-isolation checks, supervised permanent-path shore proof, individual charge/rest/ <=0.1V matching and parallel-bank closeout, AC-out branch/GFCI commissioning, secondary-alternator commissioning, final Cerbo mounting, and board strain-relief/abrasion-control.

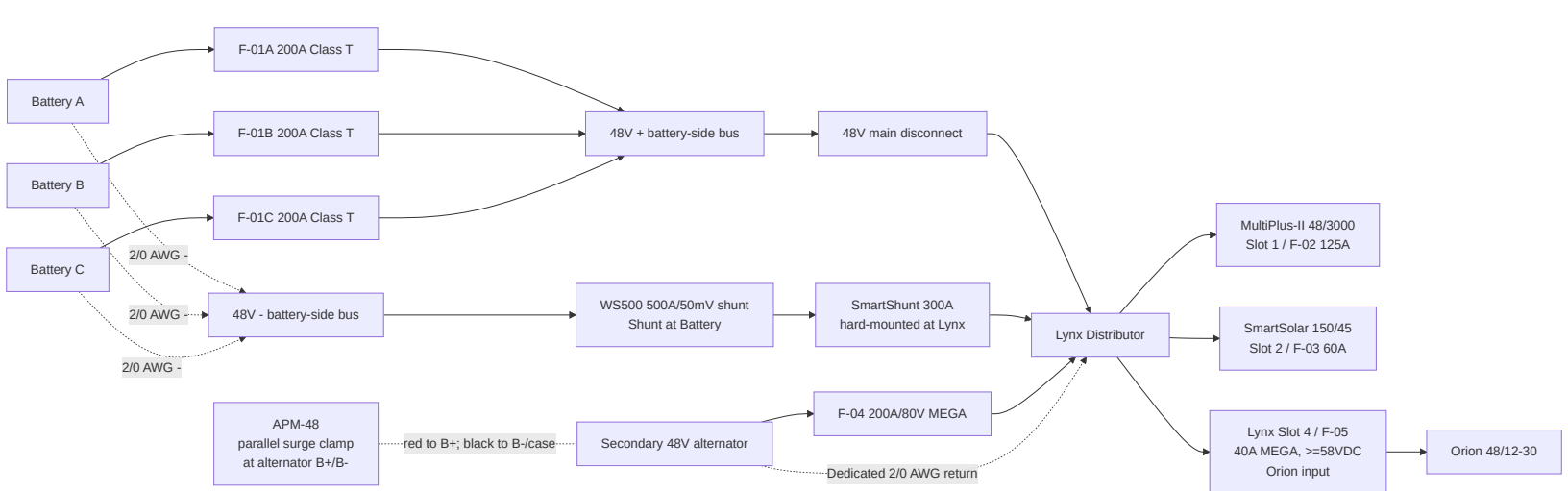
Locked component set

Function	Locked baseline	BOM row(s)
House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6
Battery branch protection	F-01A/B/C 200A Class T; manual-backed battery limit is 200A max continuous discharge per battery, with terminal/manufacturer fuse guidance still not separately specified; owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30; fed directly from Lynx Slot 4 through one verified 40A MEGA (58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) into the existing 6 AWG input pair; no separate inline/DIN input fuse; keep 6 AWG and F-07 60A/80V on 12V output	20, 10, 11
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Solar controller	Purchased Victron SmartSolar MPPT 150/45; retain it for the final array	25
Solar array	Purchased 4x Renogy 175W flexible monocrystalline = 700W; current candidate is one 4S string at 78.0V Vmp, 95.6V Voc, 8.98A Imp, and 9.50A Isc. Published -0.31%/C Voc coefficient gives 114.86V nominal calculated Voc at -40C. Exact received labels and hot-start/tracking behavior remain acceptance gates.	24
Alternator kit	Mechman 48V secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 200A/80V MEGA in Lynx Slot 3	170
WS500 PH-VAN and shunt sensing	One Eaton/Bussmann HEB-AA in-line holder + Littelfuse KLKD015.T (15A, 600VAC/DC) midget fuse on the short red combined regulator-power/positive-sense lead; separate 500A/50mV Wakespeed shunt on the battery side of the hard-mounted Victron SmartShunt in the common battery-negative path	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

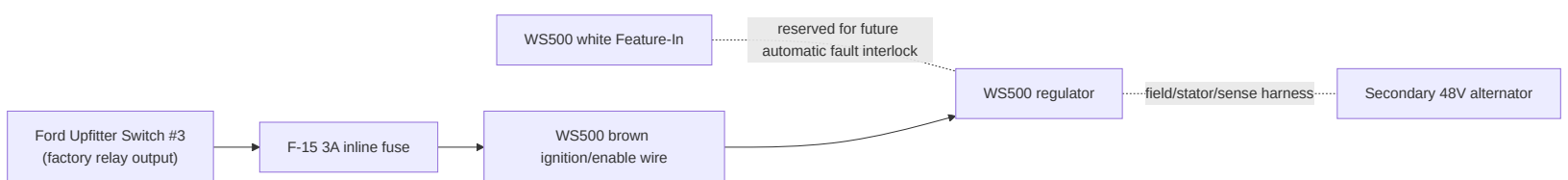
Battery manual limits and charger-programming baseline

- Battery model basis: Dumfume 51.2V 100Ah LiFePO4, 3x in parallel (1S3P ; manual allows up to 1S4P).
- Per-battery charge voltage: 58.4V +/-0.2V ; charge-limit/over-charge protection voltage also listed as 58.4V .
- Per-battery current references: recommended charge 20A ; max continuous charge 100A ; recommended discharge 50A ; max continuous discharge 200A .
- Bank-level reference for current 3x setup: recommended charge 60A ; max continuous charge 300A ; recommended discharge 150A ; max continuous discharge 600A .
- Protection thresholds: over-discharge protection 36.8V , recovery 43.2V ; discharge overcurrent protection 600A ; short-circuit protection 1800A .
- Temperature thresholds: low-temp charge protection approx 41F-50F with recovery at 50F ; high-temp protection listed as 157F / 140F .
- MultiPlus-II charger limit is 35A , so full-output charging is below the bank's 60A recommended-charge reference (~11.7A per battery in 3P). For first household-outlet tests, keep AC input current at 10A , then 12A max on a normal 15A source. Current charger-programming target is 56.8V absorption/charge, 54.0V float, 52.8V storage if shown, 0.5h max absorption, equalization off, and temperature compensation off.

48V power path



Alternator control path



Control logic

- Upfitter #3 ON : WS500 brown wire is energized, regulator is allowed to run, alternator charging can occur.
- Upfitter #3 OFF : WS500 is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use Upfitter #3 as the manual alternator-charge shutdown control.
- Do not use the main 48V disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why Upfitter #3

- It is already a factory-switched, relay-driven 12V control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
F-01A/B/C	Battery branch positive protection	200A Class T provisional	2/0 AWG
F-04	Alternator branch into Lynx Slot 3	200A/80V MEGA	2/0 AWG
F-05	Lynx Slot 4 -> Orion 48V input	MEGA 40A, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback	Existing 6 AWG direct to Orion; single input fuse
F-06	Retired standalone Orion input fuse position	Not installed	MIDI/FKS/DIN concepts superseded; do not stack with F-05
F-12/F-13-PHVAN	WS500 PH-VAN combined regulator-power / positive-voltage-sense red lead	Littelfuse KLKD015.T, 15A fast-acting midget fuse, 600VAC/DC, in Eaton/Bussmann HEB-AA 600V in-line holder	In-line beside F-04 load/alternator-side stud; short 14 AWG pigtails; sealed step-down splice to 16 AWG harness; no panel, DIN rail, or enclosure

CERB0-PWR	Cerbo GX low-current power feed	1A-3A inline fuse/holder rated for the 48v bank maximum; system/load side of main disconnect preferred for bench shutdown	18 AWG red/black duplex acceptable
WS500 current-sense pair	Purple/grey to Wakespeed shunt in common battery-negative path	No fuse; purple/high to battery-negative side, grey/low to SmartShunt/Lynx system side; twist pair and route away from noise; configure Shunt at Battery	harness lead
F-15	Upfitter #3 to WS500 brown ignition wire	3A inline ATO/ATC on 12V control circuit	16 AWG TXL/GXL control wire

Major conductors

- Battery branch and main 48V trunk wiring: 2/0 AWG .
- Parallel battery-current sharing target is similar **total loop resistance** per battery path, not equal positive-only length. The current bench layout may use short/medium/long positive leads balanced by long/medium/short negative leads; do not add unnecessary cable coils solely to make positive leads identical.
- Secondary alternator positive path (ALT B+ -> F-04 -> Lynx): 2/0 AWG , ~20 ft one-way planning basis. The APM-48 is wired in parallel across alternator B+ and B- /case; it is not in series with this conductor.
- Secondary alternator dedicated negative return (ALT B-/case -> long 2/0 -> Lynx -): 2/0 AWG , ~20 ft one-way planning basis. Do not use sheet metal/chassis as the normal return.
- Common battery-negative measurement path: battery negative combine -> Wakespeed 500A/50mV shunt -> Victron SmartShunt -> Lynx/system negative . The SmartShunt remains hard-attached to the Lynx. Configure Shunt at Battery ; purple/high faces battery negative and grey/low faces the SmartShunt/Lynx system side.
- Orion 48V feeder: existing 6 AWG remains as the no-rework path from Lynx Slot 4 and is protected by F-05 40A MEGA (>=58VDC); it is electrically overkill, and flexible fine-strand 10 AWG would be adequate if ever replaced for unrelated reasons. MPPT battery leads and Orion 12V output remain 6 AWG .
- Upfitter #3 control lead to WS500 brown wire: 16 AWG TXL/GXL planning basis, ~6 ft one-way assumed until measured.

Grounding and bonding separation

- **AC protective earth:** the MultiPlus external M6 PE lug is mandatory in this mobile installation. Bond it with at least the Victron-manual 4 mm² conductor; current build default is 10 AWG green stranded copper to a verified truck-chassis bond point. Do not use the aluminum camper shell or 80/20 as the only protective-earth conductor.
- **Exposed aluminum shell:** add a separate corrosion-compatible bonding jumper from the Hiatus shell to the chassis/equipment-ground network and verify low-resistance continuity. Mechanical shell/bed mounting contact is not accepted as proof of a durable bond.
- **AC neutral:** do not add a fixed neutral-ground bond downstream. The MultiPlus internal ground relay remains enabled: AC-out neutral bonds to chassis in inverter mode and that bond opens when external AC is accepted.
- **DC negatives:** do not jumper MultiPlus PE/case to Lynx negative or the 12V negative bus. Keep normal 48V / 12V current on dedicated returns. The Mechman branch uses a dedicated 2/0 negative directly to Lynx/system negative. Both monitoring shunts remain in series in the one common battery-negative path, so all battery current must pass through battery -> Wakespeed shunt -> SmartShunt -> Lynx/system regardless of alternator case/chassis behavior. The SmartShunt remains directly attached to the Lynx; prove no return is landed on the battery side of either shunt before charging.

APM-48 wiring intent

- Mount the APM-48 at the alternator end of the 48V branch, as close to the alternator output/ground points as practical.
- Treat the APM-48 as a parallel surge/load-dump clamp across the alternator, not a series device in the alternator charge-current path.
- Connect APM red to alternator B+ .
- Connect APM black to alternator B- , or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do **not** place either APM connector under the main battery cable lugs; Balmar's quick-start sheet explicitly prohibits placing APM connectors under the battery cable lugs.

Normal operating sequence

1. Main 48V disconnect closed.
2. Engine running.
3. Upfitter #3 switched ON .
4. WS500 enabled and regulating.
5. Alternator charges the house bank through APM-48 and F-04 .

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch Upfitter #3 OFF to disable the WS500 .
2. Wait for alternator charge current to collapse.
3. Open the main 48V disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The APM-48 is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the 48V bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- F-04 200A/80V MEGA stock and the Eaton/Bussmann HEB-AA + Littelfuse KLKD015.T in-line PH-VAN branch package were ordered 2026-08-09 ; receipt, body-marking, physical fit, and small termination-stock checks remain open. Procurement/physical inventory confirmation for the separate F-15 3A Upfitter-control hardware also remains open.
- Physical fit, stud-size, and jumper-length confirmation for the Wakespeed shunt inserted on the battery side of the SmartShunt in the common battery-negative path; the SmartShunt-to-Lynx connection does not move.
- Exact Mechman field-voltage/WS500 derate setting and alternator negative/case-isolation behavior in the installed kit.
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the WS500 .
- Whether future automatic fault-interlock logic should be added on the WS500 Feature-In or through an external relay.
- Solar purchase release: final MaxxAir/track survey, 1:1 panel/Starlink templates, two-panel cassette design, preliminary itemized <=75 lb budget, and any required insurer/Hiatus acceptance. Product/topology are locked; purchase is not yet released.
- Solar installation/travel release: received labels/polarity, built-cassette load/uplift proof, actual complete <=75 lb weigh-in, branch/disconnect/cable/umbilical/entry hardware, bond-coupon/removal proof, estimated hot-string voltage minus measured route drop, and stationary hot-roof commissioning.

Rule for future edits

- Update this file first for any 48V architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.